

DATE	30 September 2026
TEAM ID	SWUID2026349965
PROJECT NAME	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Phase 04: Project Planning

1. Plan Overview

The project is organised into five milestones that follow the structure of the project brief. Each milestone has defined activities and a clear deliverable.

Milestone	Focus	Deliverable
1	Gemini AI initialization	Working API key with verified text and image calls
2	Core functionalities	Prompts, recommendation functions, authentication routes, project structure
3	FastAPI backend integration	All planner endpoints, history, sessions and startup tasks
4	UI development	HTML, CSS and Jinja2 pages for every route
5	Testing and optimization	Tested, validated and refined application

1.1 Work Breakdown

- **Setup:** environment, Google AI Studio account, API key, project folders.
- **AI layer:** prompt templates, JSON parsing, image input, currency handling.
- **Backend:** authentication, planners, history, sessions, CORS, static files.
- **Frontend:** landing, authentication pages, dashboard, planner forms, result views, history.
- **Quality:** testing, bug fixing, prompt tuning, fallback recommendations.
- **Delivery:** documentation and demonstration.

2. Schedule

Week	Goal	Activities	Output
1	Foundation	Brainstorming, requirements, design; Google AI Studio setup and API validation	Approved requirements, tested API
2	Core AI	FastAPI initialization, libraries, environment file, prompt functions for the three planners	Working recommendation functions
3	Backend	Authentication, tokens, sessions, history, startup task	Complete API
4	User interface	Templates, planner forms, result cards, dashboard, history page	Full interface
5	Quality	End-to-end tests, prompt tuning, validation, fallbacks	Tested build
6	Delivery	Documentation, demonstration rehearsal, final submission	Documents and demonstration

2.1 Timeline (weeks)

Task	W1	W2	W3	W4	W5	W6
Brainstorming, requirements, design	X					
Gemini setup and validation	X	X				
Prompts and recommendation logic		X	X			
Authentication, sessions, planner APIs			X	X		
User interface			X	X	X	
Testing and optimization					X	X
Documentation and demonstration					X	X

3. Roles, Tools and Resources

3.1 Team Roles

Role	Responsibilities
Project lead	Planning, coordination and mentor reviews
AI and prompt engineer	Gemini setup, prompt templates, response structure, quality checks
Backend developer	FastAPI routes, authentication, sessions, history, image upload
Frontend developer	Templates, styling, JavaScript display of results
Tester and documenter	Test cases, bug tracking, documents, demonstration script

For a small team or an individual project, one person can fill several roles; the responsibilities remain the same.

3.2 Tools

Need	Tool
Code editor	Visual Studio Code
Language	Python 3.10 or newer with a virtual environment
AI access	Google AI Studio (Get API key)
API testing	FastAPI interactive docs and browser
Version control	Git and GitHub

3.3 Cost Considerations

- Development uses the Google AI Studio free tier; usage is monitored under Usage and Billing.
- The demonstration runs locally on port 8000.
- Cloud deployment on AWS is optional and would add hosting and storage costs.

4. Risk and Quality Plan

Risk	Likelihood	Impact	Response
Gemini quota or model change	Medium	High	Keep the model name configurable; catch errors; show a fallback plan
Malformed AI output	Medium	Medium	Robust JSON extraction and one retry
Budget exceeded in AI answer	Medium	High	Strict prompt; backend recalculates the remaining budget
Schedule delay	Medium	Medium	Deliver the three planners first, polish later
Secrets stored in code	Low	High	Use the environment file and review before every commit
Broken shopping links	Medium	Low	Use search links and check them periodically

4.1 Quality Plan

- Each milestone ends with a review and demonstration to the mentor.
- Definition of done: the feature works end to end, handles errors and is documented.
- Bugs are logged with steps to reproduce, expected result and actual result.

4.2 Communication Plan

Activity	Frequency	Channel
Team meeting	Twice a week	Chat or call
Mentor review	End of each milestone	SmartInternz session
Progress report	Weekly	Shared document

5. Deliverables and Acceptance Criteria

Milestone	Deliverable	Acceptance criteria
1	Gemini 2.5 access	API key works; a text prompt and an image prompt both return valid answers
2	Core recommendation logic	Home, Party and Jewelry functions return structured plans with search terms
3	Backend	All routes respond correctly; protected routes reject users who are not logged in; history is saved
4	Frontend	Every page renders correctly on desktop and phone; results are shown in readable cards and tables
5	Tested application	All planned test cases pass; plans stay within budget; errors show friendly messages

5.1 Dependencies Between Activities

- Prompt design depends on the requirements and the response structure defined in the design phase.
- Planner endpoints depend on the recommendation functions and on authentication.
- Result pages depend on the final structure of the AI response.
- Testing depends on a working frontend and backend.

5.2 Assumptions

- The Gemini 2.5 free tier is sufficient for development and demonstration.
- Team members are available for the meetings listed in the communication plan.
- Shopping platforms keep their public search URL formats unchanged.

5.3 Conclusion

This plan divides the work into five milestones over six weeks, assigns clear responsibilities and lists the risks and acceptance criteria. Following it keeps development focused on the three planners, the secure login and the history feature.